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COURSE: **CHE 312: Transport Phenomena II**

Fall Semester 2025

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Quiz #2

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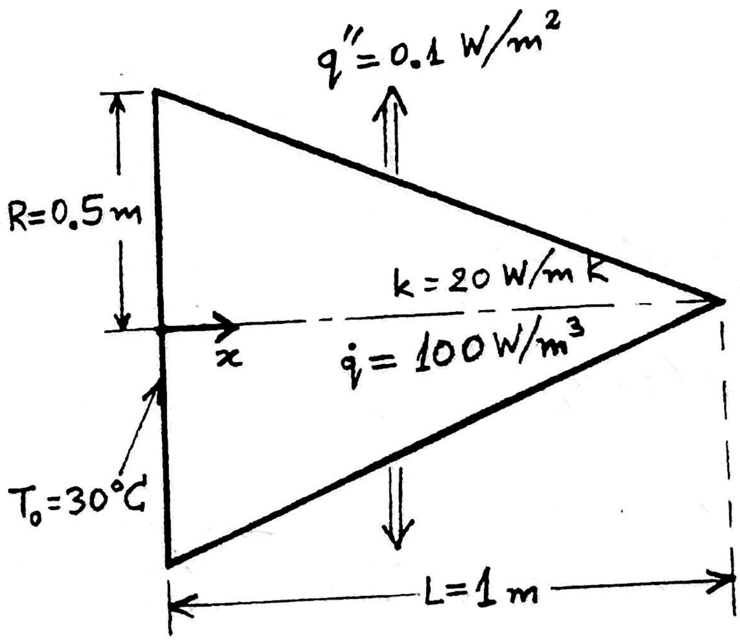
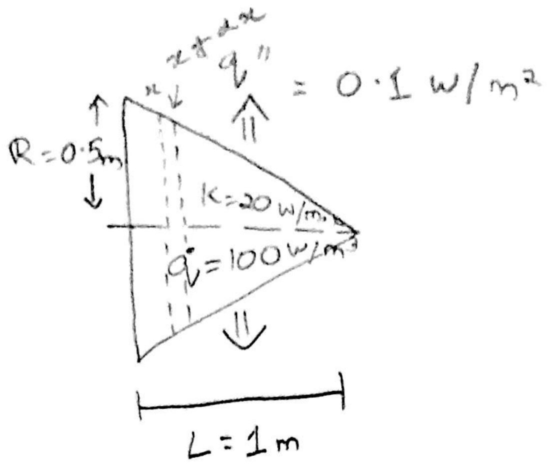
Open Notes and Book

Duration: 40 minutes

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity 20 W/m·K has uniform volumetric heat generation of 100 W/m^3. A constant heat flux of 0.1 W/m^2 is leaving the cone through the lateral surface of the cone, while the end surface at x = 0 is maintained at T_o = 30 °C. Determine the temperature profile T(x) and the temperature T(L).





Given:

$R = 0.5m$

$L = 1\ m$

$K = 20\ w/m\cdot k$

$\dot{q} = 100\ w/m^3$

$q'' = 0.1\ w/m^2$

$T(x) = ?$

$T(L) = ?$

$T_0 = 30^\circ c$

Balance eqn:

in - out + gen - removal = 0

$A(x) = \pi r^2 = \pi R^2 \left(1 - \frac{x}{L}\right)^2$

$r(x) = R\left(1 - \frac{x}{L}\right)$

Area of the thin slice: $dS = 2\pi r \sqrt{1 + \left(\frac{dr}{dx}\right)^2}\, dx$,

$\frac{dr}{dx} = -\frac{R}{L}$

$$\frac{ds}{dx} = 2\pi r s, \qquad s \equiv \sqrt{1+\left(\frac{R}{L}\right)^2} = \sqrt{\dots}$$

Assumption:

- Steady state
- constant k
- uniform volumetric generation $\dot{q}$
- uniform outward heat flux q''

$$\Rightarrow \frac{d}{dx}\left(-kA\frac{dT}{dx}\right) + \dot{q}A - q''\frac{ds}{dx} = 0$$

$$\Rightarrow \frac{d}{dx}(kAT') + \dot{q}A - q''\frac{ds}{dx} = 0$$

$$\Rightarrow k\frac{d}{dx}(\pi r^2 T') + \dot{q}\pi r^2 - 2\pi q'' r s = 0$$

$$\Rightarrow k\frac{d}{dx}(r^2T') + \dot{q}r^2 - 2q'' r s = 0$$

$$\Rightarrow \frac{d}{dx}(r^2T') = (r^2)T'' + 2(r r')T'$$

$$r = \frac{R(L-x)}{L}, \qquad r' = -\frac{R}{L}$$

$$r^2 = \frac{R^2(L-x)^2}{L}, \qquad 2rr' = -\frac{2R^2(L-x)}{L^2}$$

$$\Rightarrow k\left[\frac{R^2(L-x)^2}{L^2}T'' - \frac{2R^2(L-x)}{L^2}T'\right] + \dot{q}\frac{R^2(L-x)^2}{L^2} - 2q''\frac{R(L-x)}{L}s = 0$$

$$\Rightarrow k\left[(L-x)^2T'' - 2(L-x)T'\right] + \dot{q}(L-x)^2 - 2q''\frac{L(L-x)}{R}s = 0$$

$$\Rightarrow k\left[(L-x)T'' - 2T'\right] + \dot{q}(L-x) - \frac{2Lq''s}{R} = 0$$

$$s = \sqrt{1+(R/L)^2} = \frac{\sqrt{L^2+R^2}}{L}$$

Let $u = L - x$, $du = -dx$, $y' = \frac{dy}{dx} = \frac{-dy}{du}$

$\Rightarrow -k\left[u\frac{dy}{du} + 2y\right] + \dot{q}u - c = 0$

$\Rightarrow u\frac{dy}{du} + 2y = \frac{\dot{q}}{k}u - \frac{c}{k}$

$y(x) = T'(x)$

$\Rightarrow \frac{dy}{du} + \frac{2}{u}y = \frac{\dot{q}}{k} - \frac{c_1}{k\,u}$ $\qquad \mu(u) = \exp\left(\int \frac{2}{u}\,du\right) = u^2$

$\Rightarrow \frac{d}{du}(u^2 y) = u^2\left(\frac{\dot{q}}{k} - \frac{c}{k}\frac{1}{u}\right) = \frac{\dot{q}}{k}u^2 - \frac{c}{k}u$

$\Rightarrow u^2 y = \frac{\dot{q}}{k}\frac{u^3}{3} - \frac{c}{k}\frac{u^2}{2} + c_1$

$\Rightarrow y(u) = \frac{\dot{q}}{3k}u - \frac{c}{2k} + \frac{c_1}{u^2}$ $\qquad u = L - x$

$\Rightarrow T'(x) = \frac{\dot{q}}{3k}(L-x) - \frac{c}{2k} + \frac{c_1}{(L-x^2)}$

$x \to L$, $c_1 = 0$

$T'(x) = \frac{\dot{q}}{3k}(L-x) - \frac{c}{2k}$

$T(x) = \int T'(x)\,dx = \frac{\dot{q}}{3k}\left(Lx - \frac{x^2}{2}\right) - \frac{c}{2k}x + c_2$

BC:

$T(0) = T_0 \Rightarrow c_2 = T_0$

$c = \frac{2Lq''}{R}\sqrt{1 + (R/L)^2}$

$T(x) = T_0 - \frac{\dot{q}}{6k}x^2 + \left(\frac{L\dot{q}}{3k} - \frac{q''}{k}\frac{\sqrt{L^2+R^2}}{R}\right)x$

$T(L) = T_0 + \dot{q}\frac{L^2}{6k} - \frac{q''L}{k}\frac{\sqrt{L^2+R^2}}{R}$

$T(L) = 30 - \frac{100}{(6)(20)}(x^2) + \left(\frac{100}{3\cdot 20} - \frac{0.1}{20}\frac{\sqrt{1^2+0.5^2}}{0.5}\right)x$

$= 30 - 0.83x^2 + 1.65x \Rightarrow x = 30.82 \Rightarrow T(L) = 30.82\,^\circ C$